

project: main\_m.c, touchpanel.c, rmod.c and lcdkbd.c.

The program starts with the main function that is used to configure the various ports of ATmega8 microcon-

troller as input/output and initialise UART module, and it calls UART\_init() function to initialise to 4800bps and enable reception. The main function also enables the power-on relay

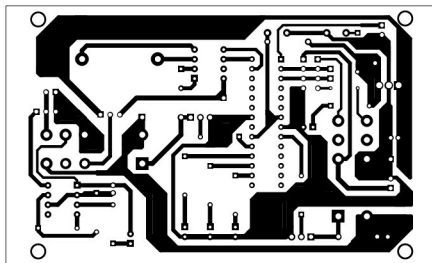


Fig. 7: Actual-size PCB of the main module

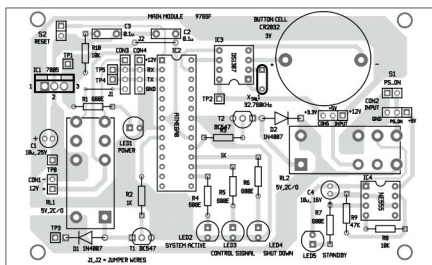


Fig. 8: Component layout of the main module

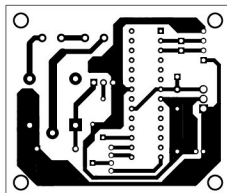


Fig. 9: Actual-size PCB of the relay module

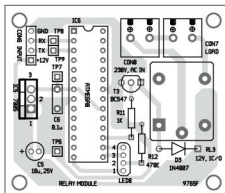


Fig. 10: Component layout of the relay module

## PARTS LIST

### Semiconductors:

- IC1, IC5, IC7, IC9 - 7805, 5V regulator
- IC2, IC6, IC8, IC10 - ATmega8 CMOS 8-bit microcontroller
- IC3 - DS1307 real-time clock chip
- IC4 - NE555 timer
- D1-D3 - 1N4007 rectifier diode
- LED1-LED7 - 5mm red LED
- LED8 - 4-pin tri-colour (RGB) LED
- LCD1 - 16x2 LCD JHD162A
- LCD2 - 128x64 GLCD JHD12864E
- T1-T3 - BC547 npn transistor

### Resistors (all 1/4-watt, 45% carbon):

- R1, R4-R7, R13, R14 - 680-ohm
- R2, R3, R11 - 1-kilo-ohm
- R8, R10 - 10-kilo-ohm
- R9 - 47-kilo-ohm
- R12 - 470-ohm
- R15 - 100-ohm
- VR1-VR2 - 10-kilo-ohm-preset

### Capacitors:

- C1, C5, C7, C9 - 10µF, 25V electrolytic
- C2, C3, C6, C8, C10 - 0.1µF ceramic
- C4 - 10µF, 16V electrolytic

### Miscellaneous:

- S1 - On/off switch
- S2 - Tactile switch
- X<sub>crystal</sub>1 - 32.768kHz crystal
- CON1 - 2-pin connector
- CON7, CON8 - 2-pin terminal connector
- CON2, CON5 - 3-pin connector
- CON3, CON4, CON6, CON9, CON11, CON12 - 4-pin connector
- CON10 - 4-pin USB-A type connector
- RL1-RL2 - 5V, 3C/O relay
- RL3 - 12V, 1C/O relay
- BATT.1 - 3V button cell
- Holder for button cell
- CPU cabinet with power and reset buttons
- ATX SMP5 power supply
- 80mmx46mm touchscreen
- 28-pin IC base
- 8-pin IC base

used in the project to provide supply to the other sections.

devswit() is used for switching between different devices connected in the system.

shut\_down() is used to shut-down the complete system in shut-down mode and configure it in standby mode.

device\_on() is used to switch on/off the corresponding relay for different appliances.

testm() is used to test system time configuration when powered on.

time\_set() is used to set RTC time and date.

d\_fnc( uint8\_t l ) is used to select various modes of operation of the system.